5.1 The natural and human causes of climate change

This chapter will explain:

★ the evidence for climate change
★ the factors that influence natural climate change
★ how human influence affects climate change.

The evidence for climate change

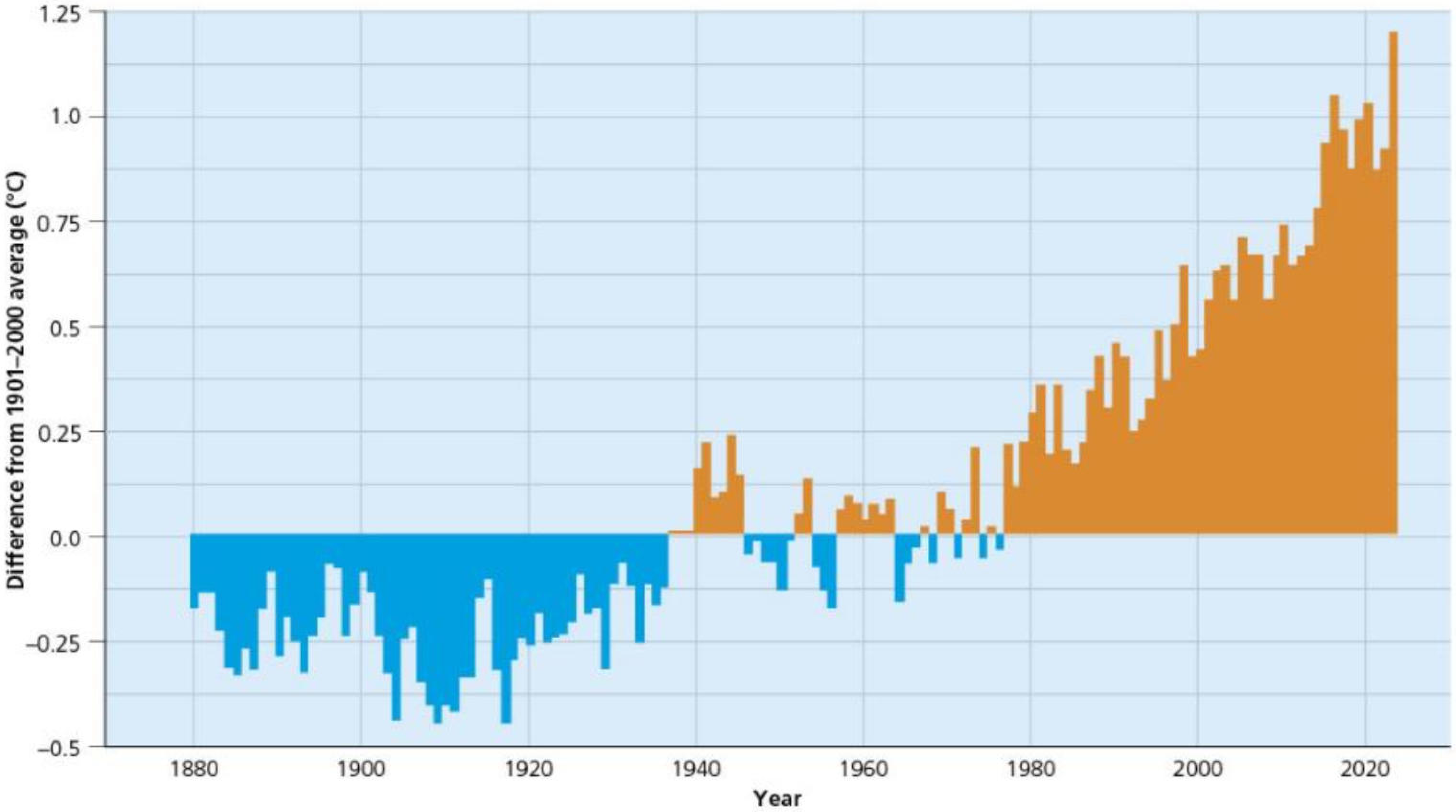
There is a range of evidence for climate change, including temperature data, ice cores, extent of sea ice and the records provided in historic writings and paintings.

Global temperature data

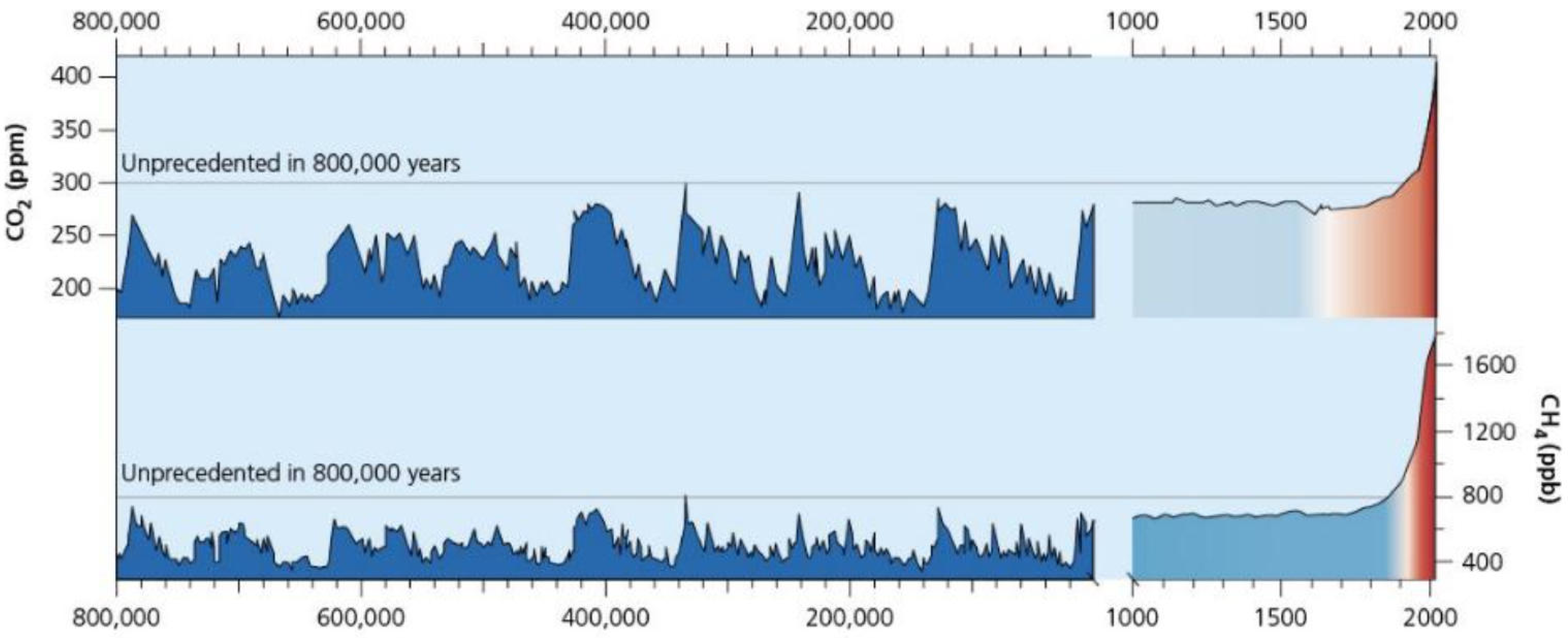
Average annual temperatures have increased by approximately 1.1°C since pre-industrial times. Figure 5.1 shows global average surface temperatures 1880–2023 compared with the average for the twentieth century. The upward trend in mean temperature is clear.

Ice cores

Ice core analysis shows that carbon dioxide (CO_2) levels, which had been steady for the previous 10,000 years at around 260–280 ppm, began to increase in the nineteenth century (Figure 5.2) and have now reached approximately 420 ppm, an increase of about 50 per cent in around 150 years. CO_2 is increasing at a rate of about 15 ppm every six years, while methane (CH_4) has doubled in the last 200 years.



▲ **Figure 5.1** Global average surface temperature 1880–2023 compared with the average for the twentieth century



▲ **Figure 5.2** Changes in atmospheric carbon dioxide (CO_2) and methane (CH_4) over the last 800,000 years, as recorded in ice cores and atmospheric sampling

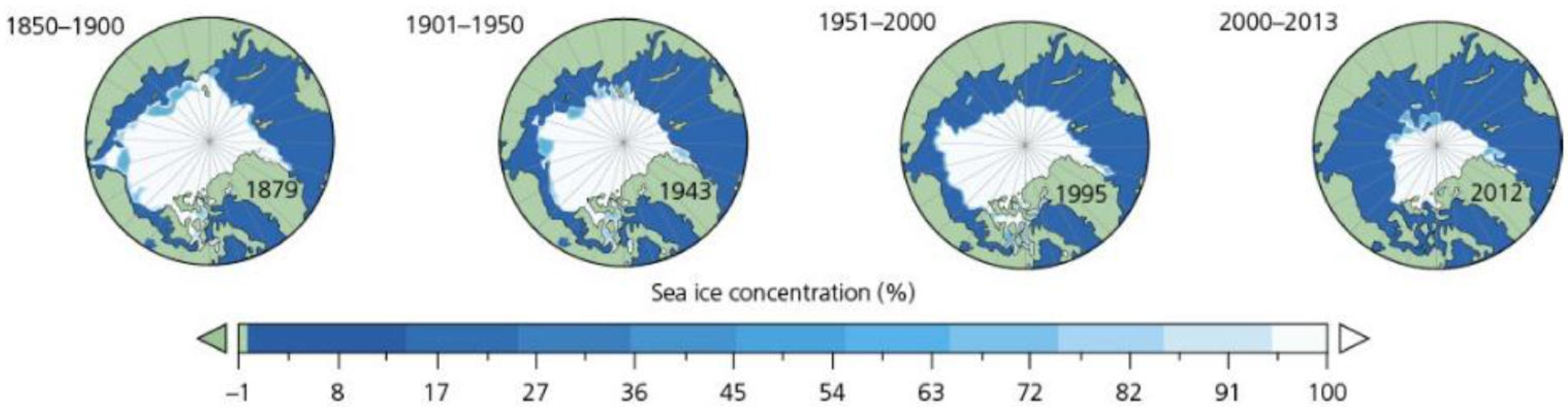
Interesting note

The history of climate change science is believed to have started with Eunice Newton Foote, an American scientist, inventor and women's rights campaigner. In 1856, she published a paper demonstrating the effects of CO_2 on temperature, in which she concluded: 'An atmosphere of that gas would give our Earth a higher temperature'.

Sea ice

Ice sheets, such as the West Antarctica ice sheet, are shrinking and thinning. Many of the world's glaciers are retreating and many smaller ones have disappeared. Snow cover in the Northern Hemisphere has decreased over the last 50 years and snow is melting earlier.

Levels of sea ice have been declining since the 1850s and the rate of this has accelerated. In each of the maps in Figure 5.3, the minimum amount of sea ice was towards the end of the period.



▲ **Figure 5.3** Sea ice cover maps for the annual minimum in September, for the periods 1850–1900, 1901–1950, 1951–2000 and 2001–2013

The decline of sea ice leads to two changes:

1. Sea levels rise.
2. There is a change in albedo as the light-coloured reflective surface of the sea ice is replaced by a darker ocean surface, which is less reflective and therefore absorbs more solar energy, reinforcing the heating process.

Sea levels have risen by about 16 cm since 1901, due to the expansion of water as it warms (the **steric effect**) and to melting ice caps and ice sheets. The rate at which sea levels rose between 2000 and 2020 was nearly double the rate of the twentieth century. In addition, the oceans are becoming more acidic as they take in some 20–30 per cent of total anthropogenic (human-made) carbon dioxide emissions. Since the Industrial Revolution, oceanic acidity has increased by 30 per cent.

Interesting note

The decline in snow and ice cover is having a major impact on species adapted to cold conditions, for example polar bears and Arctic hares. Many species may need to move to higher latitudes or altitudes, but the options for those already inhabiting high-altitude or high-latitude environments are limited. The same is true in oceans – some lower-latitude species such as tuna, cod and Alaska pollock are moving into higher-latitude oceans.

Historic writing and paintings

Some historic writings and paintings give clues about past climate. The first eight years of Charles Dickens' life (born 1812) had white Christmases (when snow falls on 25 December), and he refers to white Christmases in several of his books, including *The Pickwick Papers* (1836) and *A Christmas Carol* (1843).

Pieter Bruegel the Elder painted *The Hunters in the Snow* (1565) (Figure 5.4) following the harsh European winters of 1564 and 1565. Some people have wondered if *The Hunters* was a record of the start of the Little Ice Age.

We can conclude from these and other writings and art that there were colder periods during the past, and that these conditions were very different from the climate change that we are now experiencing.

▲ **Figure 5.4** *The Hunters in the Snow*, by Pieter Bruegel the Elder

Activities

1 Suggest how the painting *The Hunters in the Snow* by Pieter Bruegel the Elder and the writings of Charles Dickens suggest change in climate in Europe from the sixteenth century to the mid-nineteenth century to the present.

2 Describe the changes in Arctic sea ice between the periods 1850–1900 and 2000–13 shown in Figure 5.3 (page 111).

The natural causes of climate change

Orbital changes

The Quaternary Period refers to the most recent 2.6 million years of the Earth's history, up to and including the present day. The Quaternary has been characterised by frequent fluctuations of warming and cooling of the Earth, the growth and retreat of major ice sheets and ice caps, rises and falls of sea levels, the extinction of large mammals and the evolution of humans.

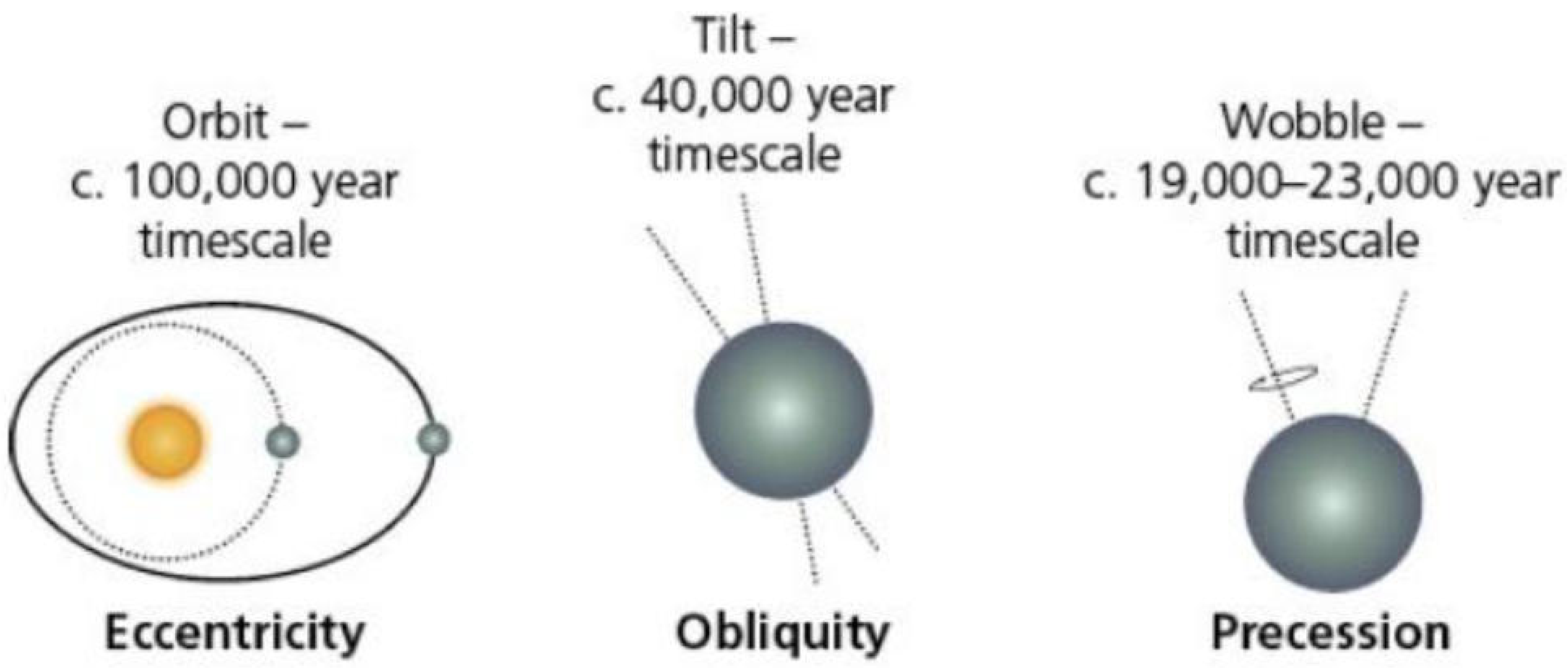
The Quaternary is a very short part of Earth's geological history. It is split into two parts, the Pleistocene Era and the Holocene Era:

- The **Pleistocene** refers to the geological era that lasted from about 2.6 million years ago to about 11,700 years ago. It was characterised by the growth and decline of ice sheets and ice caps.
- The **Holocene** refers to the most recent 11,700 years, when the Earth's climate has been relatively warm and the major ice sheets have retreated.

Nevertheless, the Earth is still in an ice age, as evidenced by the ice in Antarctica and Greenland. However, we are in a warm phase, known as an **interglacial period**. A cold phase, when ice grows, is referred to as a **glacial period**. Now, approximately 10 per cent of the Earth's surface is covered by ice. During the glacial periods, about 30 per cent of the Earth's surface is covered by ice.

Milankovitch cycles

One of the most important theories to help explain the natural causes of climate change in the Quaternary Period is that put forward by Milutin Milankovitch. The Milankovitch cycle theory (Figure 5.5) states that the amount of solar energy reaching the Earth varies with changes in the Earth's orbit, its tilt and its 'wobble' – that is, the direction of its rotation. The Earth's orbit varies over a timescale of about 100,000 years. When the Earth is further from the sun, it receives less energy. Also, when the tilt is greater, the seasons are longer. The 'wobble' determines which hemisphere – Northern or Southern – is facing the sun. This varies over a 19,000–23,000-year timescale.



▲ **Figure 5.5** The Milankovitch cycles

At the start of the Quaternary Period, the continents were largely in the positions that they are today. Since then, however, the Earth has wobbled in its rotation around the sun. These 'wobbles' have been partly responsible for the growth and decline of glaciations. By around 800,000 years ago a pattern had emerged: cold glacial phases that lasted up to 100,000 years, followed by a warmer interglacial period lasting about 10,000 to 15,000 years.

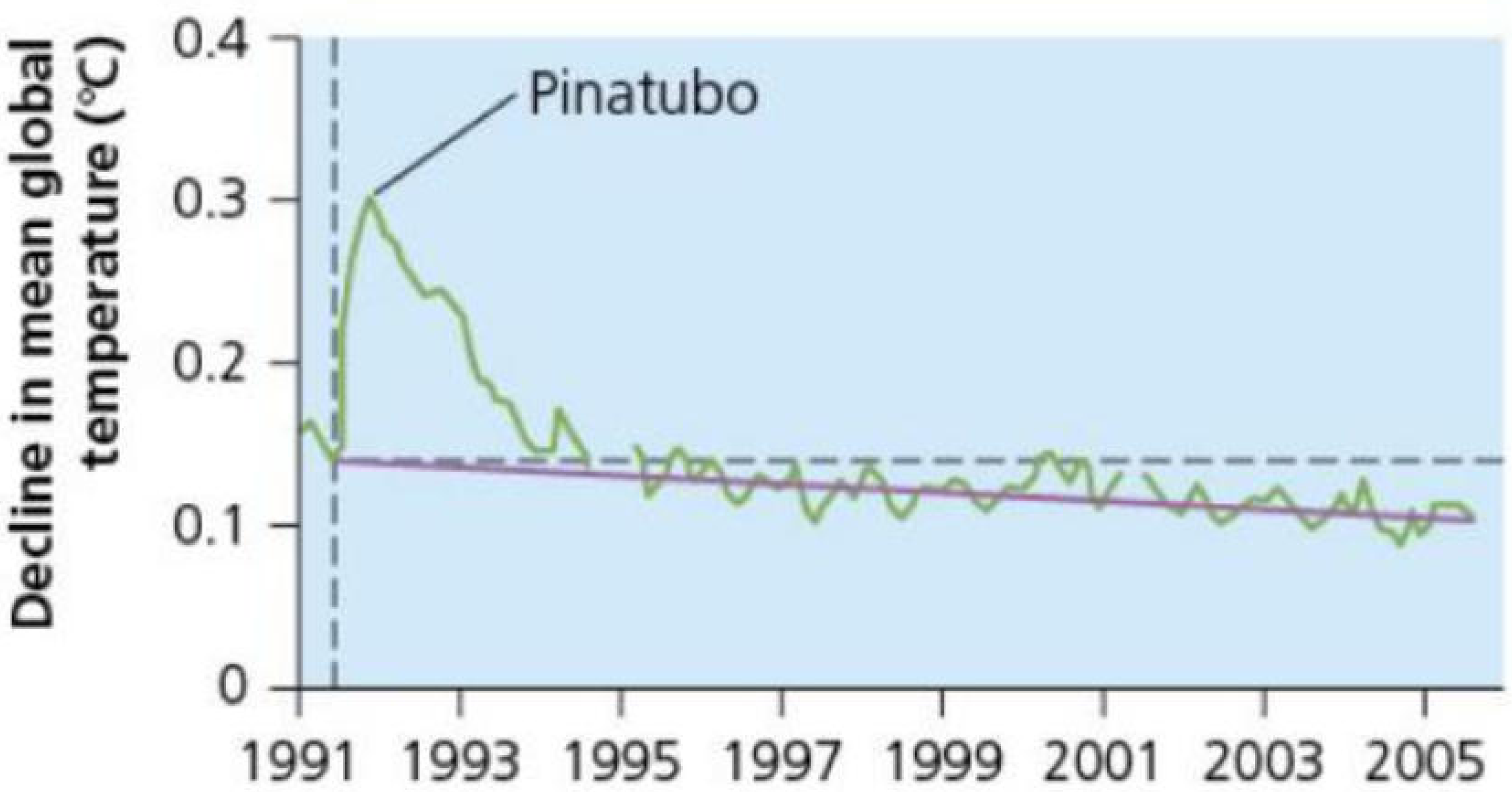
Sunspots

There is evidence that the sun has an 11-year cycle in which it brightens and dims – although the output only varies by around 0.1%. This change may have a very small effect on surface temperatures. Between 1880 and 1955, solar radiation gradually increased. However, since 1955 it has been gradually decreasing. So, although global temperatures have increased, due to global warming, solar output has decreased.

Volcanic activity

Other natural causes of climate change include volcanic eruptions (Figure 5.6). Those more likely to cause changes to climate are large eruptions, especially in tropical areas. The resulting atmospheric dust is also believed to block or reflect incoming solar energy, thereby leading to lower temperatures on Earth. Low latitudes receive a greater amount of solar energy than high latitudes, which is why volcanic eruptions in low latitudes have a greater impact on climate change – they block out a higher amount of solar radiation than high-latitude eruptions.

According to the United Nations' Intergovernmental Panel on Climate Change (IPCC), long- and short-term variations in solar activity play only a very small role in altering the Earth's climate.



▲ **Figure 5.6** The impact of the 1991 Mount Pinatubo eruption (Philippines) on temperature between 1991 and 2005: the purple line shows the global decline in sun-blocking aerosols following the eruption

Activities

1. Describe what happens to the Earth during:
   a. the 100,000-year cycle
   b. the 41,000-year cycle.
2. Explain how the eruption of Mount Pinatubo affected global climate.

The human influences on climate change

The **greenhouse effect** is the process by which certain gases (greenhouse gases) allow short-wave radiation to pass through the atmosphere but trap a proportion of the outgoing long-wave radiation from the Earth. This traps heat within the atmosphere like a greenhouse and leads to a warming of the atmosphere. It a natural process and vital for life on Earth.

The most common greenhouse gas is water vapour, which accounts for about 50 per cent of the natural greenhouse. However, the gases that account for human causes of climate change are carbon dioxide (CO_2), methane (CH_4) and chlorofluorocarbons (CFCs).

The enhanced greenhouse effect

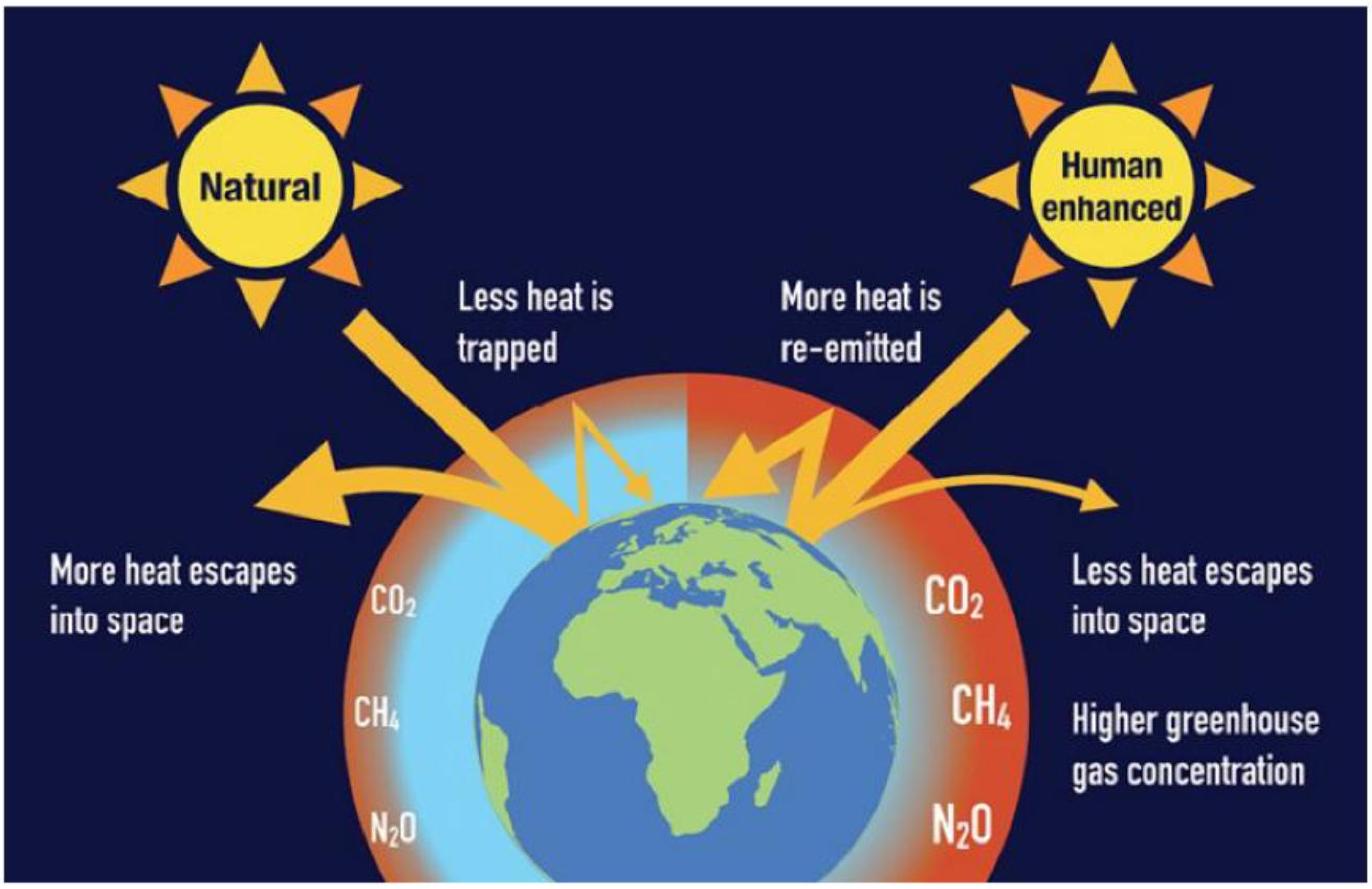
Atmospheric levels of carbon dioxide rose from around 315 parts per million (ppm) in 1950 to over 420 ppm by 2020, and they are predicted to rise to 600 ppm by 2050. This rise is due to issues such as burning **fossil fuels** (coal, oil and natural gas) and land-use changes to make way for agriculture, settlements and infrastructure. Deforestation reduces the store of carbon on the Earth (that is, in the trees) and carbon is released into the atmosphere when the trees are burned, adding to the amount of atmospheric carbon dioxide.

Agriculture is a major cause of climate change through processes including land-use change (deforestation, for example), methane emissions from cattle and rice fields, and the use of machinery in farming and in the transport of goods to market.

The **enhanced greenhouse effect** (EGHE) (Figures 5.7 and 5.8) is known by many terms, such as global climate change, climate crisis and global warming. As a result of the EGHE, temperatures have been rising since around 1850.

Activities

1 Identify three greenhouse gases.
2 Explain how greenhouse gases affect temperature on Earth.



▲ **Figure 5.7** The greenhouse effect and the enhanced greenhouse effect

▲ **Figure 5.8** Some impacts of global warming – sunbathing on glaciers and irrigation needed to play tennis